

The Pragmatics of Quantifier Scope: A Corpus Study

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Introduction

Possible Readings

- Semanticists have generally been concerned with accounting for the range of **possible scopes** for a given sentence.
- The semanticist's aim is roughly what previous literature terms *scope generation*.

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- That is, semantics is not concerned with the problem of *quantifier scope disambiguation* (QSD).

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- That is, semantics is not concerned with the problem of *quantifier scope disambiguation* (QSD).

While we agree that actual usage patterns are largely outside the domain of semantics, they are in the domain of pragmatics.

Pragmatics of quantifier scope

In order to develop a model for QSD, we examine the factors influencing quantifier scope in a controlled, but naturally occurring body of real speech: LSAT Logic Puzzles.

Goal

- Today, our aim is to introduce our corpus and report preliminary findings.

Psychologically Plausible Predictors

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Linear order/C-command

(3) Every professor saw a student. *every* >> *a*

(4) A student saw every professor. *a* >> *every*

Gillen 1991, Kutzman & McDonald 1993, Tunstall 1998, Anderson 2004

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(5) Every professor saw a student. *every* >> *a*

(6) A student saw every professor. *a* >> *every*

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Note: It is very difficult in English to separate the effect of linear order from the next predictor, grammatical function.

Psychologically Plausible Predictors

Grammatical function hierarchy

(7) Joan told a child the story at every intersection. *every* >> *a*

(8) Joan told everyone the story at an intersection. *a* >> *every*

S > Prep > IO > O

Kutzman & McDonald 1993, Tunstall 1998, Micham et al 1980.

Psychologically Plausible Predictors

Grammatical function hierarchy

(11) Joan told a child the story at every intersection. *every* >> *a*

(12) Joan told everyone the story at an intersection. *a* >> *every*

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loup's (1975) Quantifier Hierarchy

(13) She knows a solution to every problem. *every* >> *a*

(14) She knows a solution to all problems. *a* >> *all*

each > every > all > most > many > several > some_{pl} > a few

Tunstall 1998, Van Lehn (1978).

Computationally Effective Predictors

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A doctor lives in every city.

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Srinivasan & Yates (2009) show that numerical typicality can be extracted from a large corpus and applied successfully to QSD.

- Applied to a handpicked corpus of 46 items, information about numerical typicality significantly improves prediction, especially for indirect scope.

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- The result was 893 sentences, coded for scope by 2 people

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Corpus Worries

- Leave out NPs headed by *a/an*
- Do not separate conjoined or appositive clauses. One result is that the two quantifier do not interact in 61% of the corpus.

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Note: The most active features of each model were things like, intervening comma, intervening conjunct node, intervening quotation mark, intervening colon, etc.

Summary

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- Linear order/C-command
- Grammatical hierarchy
- Particular quantificational item
- Intervening clause boundaries
- World knowledge

LSAT Logic Puzzles

LSATs

- The LSAT exam consists of several types of questions: reading comprehension, analytical reasoning, etc.
- Our corpus is drawn from one particular type of question: Analytical Reasoning Questions, or Logic Puzzles.
- Logic Puzzles follow a particular format as follows:

Structure of a logic puzzle

In the course of one month Garibaldi has exactly seven different meetings. Each of her meetings is with exactly one of five foreign dignitaries: Fuentes, Matsuba, Rhee, Soleimani, or Tbahi. The following constraints govern Garibaldi's meetings:

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She has exactly three meetings with Fuentes, and exactly one with each of the other dignitaries.

She does not have any meetings in a row with Fuentes.

Her meeting with Soleimani is the very next one after her meeting with Tbahi.

Neither the first nor last of her meetings is with Matsuba.

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2. If Garibaldi's last meeting is with Rhee, then which one of the following could be true?

} Question

- (A) Garibaldi's second meeting is with Soleimani.
- (B) Garibaldi's third meeting is with Matsuba.
- (C) Garibaldi's fourth meeting is with Soleimani.
- (D) Garibaldi's fifth meeting is with Matsuba.
- (E) Garibaldi's sixth meeting is with Soleimani.

} Answers

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Why logic puzzles?

Minimal Ambiguity

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- As an aptitude test, the LSAT explicitly states assumptions which might be left to world knowledge in ordinary conversation.
- In essence, the entire discourse context is made linguistically explicit, allowing us to extract away from world knowledge.

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Multiple quantifiers frequent

- Sentences with two or more quantifiers are, unsurprisingly, quite frequent in this register.

Scopal Domains

Syntactic Constraints

- In Higgins & Sadock (2003), the sentence was taken as the domain for quantifier scope regardless of syntactic complexity.
- However, it is often clear that a sentence consists of multiple separate scopal domains.
- For example, if two quantifiers appear in a coordinate structure as in (??)

(15) Joe ate three oranges and Pam ate two apples.

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- However, it is often clear that a sentence consists of multiple separate scopal domains.
- For example, if two quantifiers appear in a coordinate structure as in (??)

(17) [Joe ate three oranges]₁ and [Pam ate two apples]₂.

- The example, then is best treated as two separate scopal domains, one per conjunct.

Scopal Domains (cont'd)

- Quotations and Parenthetical content like appositive relative clauses similarly involve multiple scopal domains.
- Therefore, we consider *scopal domains* with multiple quantifiers, rather than *sentences*.
- This is consistent with our stated goal of studying the *pragmatics* of quantifier scope.
- The lack of relative scope between quantifiers in different conjuncts of a coordinate clause is an observation about the syntax/semantics of quantifiers, not their pragmatics.

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- The lack of relative scope between quantifiers in different conjuncts of a coordinate clause is an observation about the syntax/semantics of quantifiers, not their pragmatics.

This approach does not entirely eliminate the role of syntax/semantics in determining possible readings, but it accounts for the most common cases (cf. Higgins & Sadock (2003)'s finding that commas, 'and', etc. are the best predictors in their models)

Tagging the data

Procedure

- Quantifier scope by its nature requires a trained linguist to tag.
- First, we separated the data into individual sentences and then further into scopal domains.
- Second, we enlisted undergrads to identify sentences with multiple quantifiers and give a first pass at tagging them.
- Finally, we individually went through the corpus by hand, producing the final tags.

No effort was made to quantify inter-annotator agreement since (i) this would require additional skilled coders, and (ii) Higgins & Sadock (2003)'s study did do this, found fairly high variability, and concluded (reasonably) that this is fairly inescapable.

Categories tagged

Scope

- The relative scope of the 2 or more quantifiers in a scopal domain

3 Factors

- 1 Linear order
- 2 Syntactic position
- 3 Lexical identity of quantifier

Categories tagged (cont'd)

Scope

- We coded scope numerically, with 1 corresponding to widest scope and other numbers indicating narrow scope.
- Quantifiers with no relative scope (mainly cumulative readings) were 'co-tagged' with the same number.
- This is merely a convenience for examples with 2 quantifiers,
- ... but necessary for sentences with 3 or more, where two quantifiers may take scope relative a third quantifier, but not relative to one another.
- In cases where no truth conditional difference was clear, we used the felicity of "such that" paraphrases as our ultimate criterion.

Categories tagged (cont'd)

Linear Order

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Lexical identity

- We used a 'splitting' strategy here, having the entire complex determiner tagged in cases like 'more.than.two', 'a.different', etc.

Examples

Tagged examples

- (18) Each**&1_S_each#** tape is to be assigned to a different**&2_to_a.different#** time slot, ...
- (19) Each**&1_S_each#** professor has one or more**&2_O_one.or.more#** specialities,
- (20) . . and no**&1_S_no#** tape is longer than any other**&2_than_any.other#** tape.
- (21) The judge of the show awards exactly four**&1_O_exactly.four#** ribbons to four**&1_to_four#** of the dogs,

The dataset

- we focus on sentences with 2 quantifiers only
- 450 sentences, i.e., 900 quantifiers / observations

scope

narrow:416, wide:484

lin.ord

first:450, last:450

gram.fun

S:342, O:199, PREP.MISC:86, A:70, IN:48, P:35 etc.

lex.real / lex.real.other

each:170, card.num:158, a:139, exactly:133, no:52, at.least:38 etc.

The dataset

- we remove the cumulative sentences
 - we are left with 828 observations / quantifiers out of 900
- we focus on S and O only
 - we are left with 489 observations
- we have double counting: some sentences have both an S and an O quantifier and the scope of one completely determines the scope of the other
 - 141 doubly counted sentences
 - we randomly sample one quantifier from each of them

The S & O dataset without double counting

- 348 observations

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Response Variable

- scope: factor w/ 2 levels "narrow","wide"

2 fixed effects

- lin.ord: factor w/ 2 levels "first","last"
- gram.fun: factor w/ 2 levels "S","O"

2 random effects

- lex.real: factor w/ 19 levels "a","a.different",...
- lex.real.other: factor w/ 22 levels "a","a.different",...

The S & O dataset without double counting

scope

narrow:137, wide:211

lin.ord

first:251, last:97

gram.fun

S:235, O:113

lex.real

each:82, a:49, exactly:49, card.num:46, no:33, at.least:17 etc.

lex.real.other

a:61, each:61, exactly:58, card.num:41, at.least:18, no:16 etc.

The S & O dataset without double counting

```
lin.ord = first
```

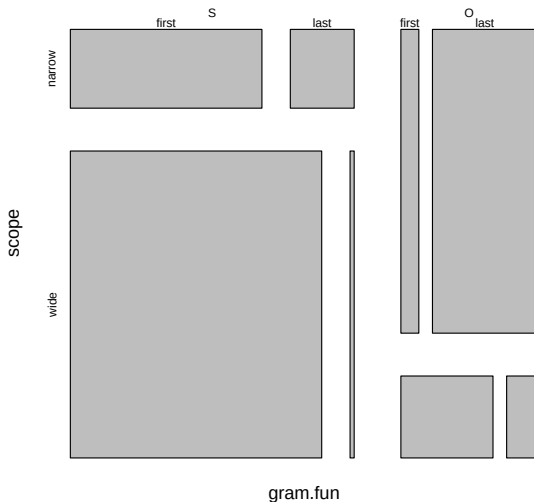
	gram.fun		
scope	S	0	
narrow	36	13	
wide	184	18	

```
lin.ord = last
```

	gram.fun		
scope	S	0	
narrow	12	76	
wide	3	6	

The S & O dataset without double counting

`xtabs(~gram.fun + scope + lin.ord, data = quant2)`



The fixed-effects logistic regression

```
glm(formula = scope ~ lin.ord + gram.fun, binomial)
```

Coefficients:

	Estimate	
(Intercept)	1.6247	
lin.ordlast	-2.9284	
gram.fun0	-1.2723	
	Std. Error	
(Intercept)	0.1778	
lin.ordlast	0.4278	
gram.fun0	0.3595	
	z value	Pr(> z)
(Intercept)	9.139	< 2e-16
lin.ordlast	-6.845	7.66e-12
gram.fun0	-3.539	0.000402

The estimated probabilities of / preferences for wide scope

gram.fun	lin.ord	wide.scope.prob
S	first	0.84
0	first	0.59
S	last	0.21
0	last	0.07

The fixed-effects logistic regression

Single term deletions

Model:

```
scope ~ lin.ord + gram.fun
```

	Df	Deviance	AIC
<none>		296.22	302.22
lin.ord	1	354.80	358.80
gram.fun	1	307.77	311.77

	LRT	Pr(Chi)
--	-----	---------

<none>

lin.ord	58.577	1.955e-14
gram.fun	11.547	0.0006784

Testing for interactions

Analysis of Deviance Table

Model 1: scope ~ lin.ord + gram.fun

Model 2: scope ~ lin.ord * gram.fun

	Resid. Df	Resid. Dev	Df
1	345	296.22	
2	344	296.19	1

	Deviance	P(> Chi)
1		
2	0.031033	0.8602

Model fit

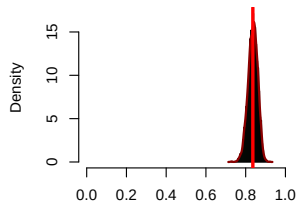
C

- C is an index of concordance between predicted probability and observed response
- C: 0.8327049

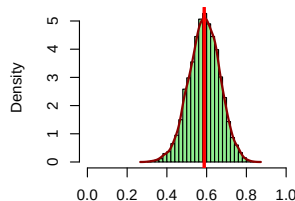
Dxy

- Somers' Dxy is a rank correlation between predicted probabilities and observed responses related to C
- Dxy: 0.6654098

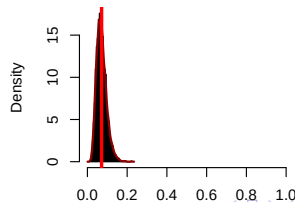
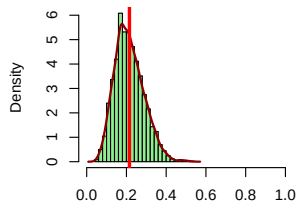
Bayesian estimation with vague priors (WinBUGS)



$p(\text{wide}|\text{S.first})=0.84$



$p(\text{wide}|\text{O.first})=0.59$



Adding random intercepts for lex.real and lex.real.other

```
scope ~ lin.ord + gram.fun + (1 | lex.real) +  
                                     (1 | lex.real.other)
```

AIC	BIC	logLik	deviance
195.1	214.3	-92.54	185.1

Random effects:

Groups	Name
lex.real.other	(Intercept)
lex.real	(Intercept)

Variance	Std.Dev.
3.2021	1.7894
1.9458	1.3949

Number of obs: 348,

groups: lex.real.other, 22; lex.real, 19

Adding random intercepts for lex.real and lex.real.other

Fixed effects:

	Estimate		
(Intercept)	2.7355		
lin.ordlast	-4.0049		
gram.fun0	-1.2992		
	Std. Error		
(Intercept)	0.7361		
lin.ordlast	0.8084		
gram.fun0	0.5598		
	z value	Pr(> z)	
(Intercept)	3.716	0.000202	
lin.ordlast	-4.954	7.26e-07	
gram.fun0	-2.321	0.020296	

Random effects: lex.real

(Intercept)

a	2.099174351
a.different	-0.506734061
any	0.845720344
at.least	-0.080778537
at.most	-0.937468611
both	0.193133614
card.num	-1.872397945
each	0.945132118
either	-0.927069451
every	0.071068196
exactly	-0.915708184
more.than	-0.227894863
neither	0.925935231
no	0.262491177
only	0.026563387
or	-0.005507181
some	-0.217679375
the.miscellaneous	0.704618957
the.same	-1.056184432

Random effects: lex.real.other

	(Intercept)
a	-1.60702550
a.different	0.46597845
all	-0.58133210
any	0.76323966
at.least	0.03052082
at.most	1.33107446
both	0.26470233
card.num	-1.00912983
each	-3.29041107
each.other	-0.32640791
either	1.63889213
every	0.30350598
exactly	0.89403236
modifier.miscellaneous	0.93729048
more.than	0.85489781
neither	-1.31125004
no	-1.53925127
only	-0.40543996
or	0.07329231
some	0.26035663
the.miscellaneous	-0.37408166
the.same	1.51805135

Random-effects logistic regression: model fit

Mixed effects logistic regression

- C: 0.9778254
- Dxy: 0.9556509

Compare with the fixed-effects logistic regression

- C: 0.8327049
- Dxy: 0.6654098

Compare with the random-effects regression w/o fixed effects

- formula: $\text{scope} \sim 1 + (1 \mid \text{lex.real}) + (1 \mid \text{lex.real.other})$
- C: 0.9624139
- Dxy: 0.9248279

Testing for interactions

```
scope ~ lin.ord + gram.fun + (1 | lex.real) +
                                     (1 | lex.real.other)
scope ~ lin.ord * gram.fun + (1 | lex.real) +
                                     (1 | lex.real.other)
```

Df	AIC	BIC
5	195.09	214.35
6	195.89	219.00

logLik	Chisq	Chi	Df
-92.543			
-91.943	1.2011		1

Pr(>Chisq)

0.2731

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First of all, we would like to thank the Law School Admission Council (LSAC) for access to practice test materials used in the analysis. We would also like to thank Pranav Anand, Donka Farkas, Matt Wagers, and participants in the UCSC Corpus Linguistics Group for helpful feedback.

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